

Lesson 5

Introduction to distributed parameter systems

1 Introduction

In the previous lessons, the dynamics of lumped parameter systems was treated. These systems are assumed to have a finite number n of degrees of freedom, which normally implies the assumption that the system is formed by a number of rigid bodies connected each other by springs, dampers and different types of constraints (hinges, clamps etc.).

In order to study mechanical systems of this type, one should be able to clearly identify bodies and parts where the inertial properties of the system are lumped, and other bodies or parts for which elastic and damping effects are predominant whereas inertial effects may be neglected. As an example, when studying the vibration of a car in the vertical plane, one could establish a simple model by considering the carbody as a rigid body connected to the ground through the front and rear suspensions, each one modelled as a spring and a damper.

However, there are mechanical systems where the separation into parts behaving (approximately) as rigid bodies and other parts that may be reduced to massless springs and dampers is not feasible. For instance, if we consider the case of the shaft of a steam turbine + generator, we recognise that this system may be modelled as a beam resting on several supports: if we want to study the bending vibrations that may be produced by an unbalance in the shaft, we necessarily need to define an approach taking into account the presence of inertial, elastic, viscous forces as “distributed” inside the whole structure, and we need then to change our approach to consider instead of a *discrete* (or *lumped parameter*) system a *continuous* (or *distributed parameter*) system.

We also note that the choice between using a lumped parameter or distributed parameter model for a system should be based on the aims of the study, particularly in terms of the frequency range (or pass band) addressed. For instance, a model of a road vehicle representing the car body as a rigid body is adequate to analyse the “low frequency” behaviour of the vehicle in terms e.g. of comfort, road holding and handling in a frequency range up to 10-15 Hz approximately. If otherwise vehicle dynamics is being investigated in a wider frequency range, e.g. when dealing with noise problems involving a frequency range up to some kHz, a more detailed model would be needed to properly account for non-rigid vibration of the car body and some other car components (doors, roof, covering panels,...).

This lesson provides an introduction to the mechanics of distributed parameter systems. Because distributed parameter systems consist of bodies undergoing non-rigid movements, we first need to introduce suitable models for deformable bodies (a topic known as ‘solid mechanics’). This entails introducing the notions of strain and stress and taking into account material’s behaviour in terms of its constitutive law.

We will confine our analysis to the simple case of small non-rigid deformation and will consider solids having linear elastic material behaviour. These assumptions are nevertheless satisfied by a quite wide variety of mechanical systems and are therefore fully adequate to the aims of this course. Also, we restrict ourselves to the study of *beams*. As will be described in more detailed in this lesson, a beam is a solid having special geometric properties, allowing to describe its deformable motion in terms of one single space coordinate instead of three ones, as is required when dealing with the case of a three-dimensional body having generic geometry.

2 What is a deformable body?

A deformable body is a body that may perform a non-rigid motion. We recall a rigid motion is a motion in which:

- i) the distance between any two points attached on the body is not changing;
- ii) the angle formed by any two segments attached on the body is not changing.

We may exemplify the kinematics of a rigid body using the simple 2-dimensional case shown in Figure 2.1: we consider a 2D solid laying in the x-y plane and we denote by:

$$\begin{aligned} u &= u(x, y) \\ v &= v(x, y) \end{aligned} \quad (2.1)$$

the horizontal and vertical components of the displacement of any generic point P in the body. The overall displacement of the body is hence defined by function u and v defined over the whole surface of the 2D body.

In the case when a rigid motion is prescribed for the body, these functions must satisfy *rigidity constraints* that may be stated as:

$$\begin{aligned} \overline{AB} &= \text{const} \text{ for any } A \text{ and } B \text{ in the body} \\ (\hat{CAB}) &= \text{const} \text{ for any } A, B \text{ and } C \text{ in the body} \end{aligned} \quad (2.2)$$

It follows from (2.2) that the displacement of any point in a rigid 2D body can be expressed as function of 3 independent kinematic parameters, e.g. the horizontal and vertical displacement of a single point A and the angle formed by a line AB in the body with a fixed direction, e.g. the x axis, which is called the *rotation* of the body. Indeed, once defined the position of point A and the inclination of the AB line, the position of any point B on that line is known thanks to the first of (2.2) while the position of any point C not belonging to the AB line is uniquely defined because the direction of the AC line comes from the second of (2.2) and the distance from A on that line comes from the first of (2.2).

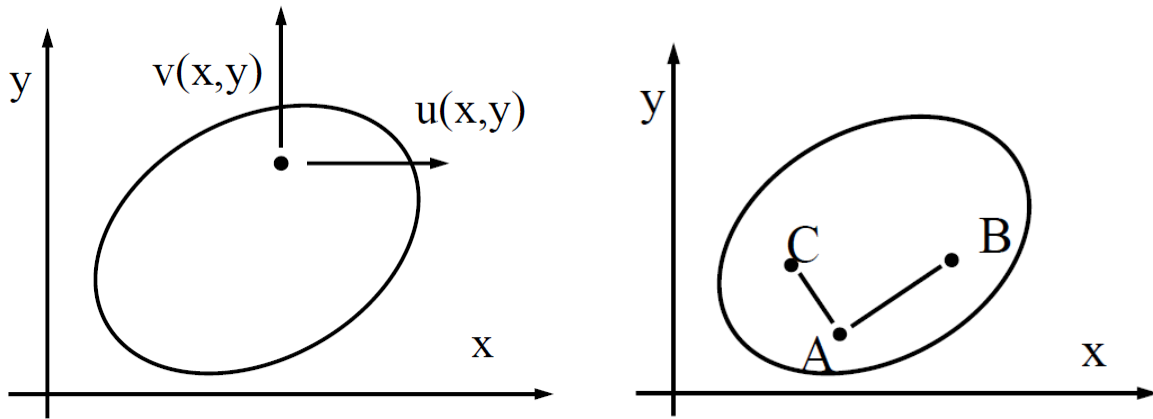


Figure 2.1: Motion of a 2D solid

Based on the above recap of rigid body motion, we recognise that in a non-rigid motion:

- i) the length of a segment $\overline{AB}$ may be changing;
- ii) the angle $(\hat{CAB})$ formed by two lines AB and AC may be changing.

in order to describe a non-rigid motion, we need therefore to define a *measure* of the changes in the length of segment and of the changes in the angle formed by the *fibres* of the body. This is best done in non-dimensional terms and leads to the definition of the strain components, which is provided for the case of small non-rigid displacements in the next section.

3 Strain components for small displacements

First of all, we restrict to the case of *small displacements*, i.e. we consider a displacement of the body whose components u and v defined by (2.1) are so small that can be considered as infinitesimal quantities. It follows that than in any mathematical treatment involving these quantities powers higher than one of u and v can be neglected.

An elegant proof is available showing that any infinitesimal displacement in the form (2.1) can be decomposed into a rigid component (which in turn may be described in terms of three independent coordinates) and a deformable component which may be defined in terms of non-dimensional parameters called the *strain components*. We prefer here to introduce the stress components in a more intuitive way, by considering some special cases of non-rigid displacement.

3.1 Normal strain

We start by considering the special case in which the body is only undergoing a displacement in horizontal direction and the displacement amount only depends on the x coordinate:

$$\begin{aligned} u &= u(x) \\ v &= 0 \end{aligned} \quad (3.1)$$

this case is shown in Figure 3.1, with reference to an infinitesimal portion of the 2D solid consisting of a rectangle ABCD with point A having generic coordinates x, y and:

$$\overline{AB} = \overline{CD} = dx$$

$$\overline{AC} = \overline{BD} = dy$$

A', B', C', D' are the position of the four vertexes of the rectangle after the displacement has taken place.

Because of the special case considered, we recognise that the fibres initially aligned with the x axis will stretch (or compress) on account of the displacement applied being non-uniform, whereas fibres aligned with the y axis will not undergo any change in their length. Furthermore, the orientation of fibres aligned along both coordinate axes is not changed by the special displacement considered.

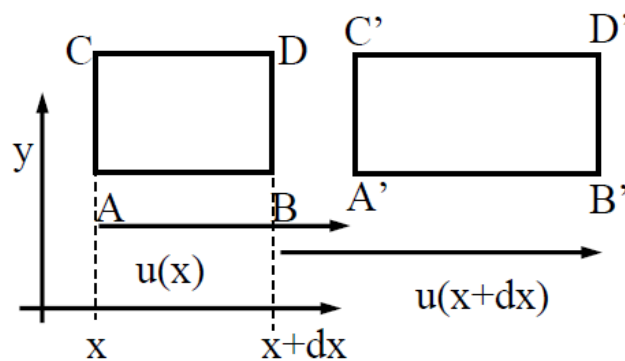


Figure 3.1: Non rigid motion in x direction only

We introduce the normal strain in x direction ε_x as a non-dimensional quantity measuring the amount by which fibres laying in the x direction are changing their length as the consequence of the non-rigid displacement:

$$\varepsilon_x = \frac{\Delta \ell_x}{\ell_x} \quad (3.2)$$

Considering (3.1), (3.2) becomes:

$$\varepsilon_x = \frac{\overline{A'B'} - \overline{AB}}{\overline{AB}} = \frac{u(x+dx) - u(x)}{dx} = \frac{u(x) + \frac{\partial u}{\partial x} dx - u(x)}{dx} = \frac{\partial u}{\partial x} \quad (3.3)$$

Next, we consider a special case of non-rigid displacement according to:

$$\begin{aligned} u &= 0 \\ v &= v(y) \end{aligned} \quad (3.4)$$

The displacement of the ABCD rectangle is shown in Figure 3.2

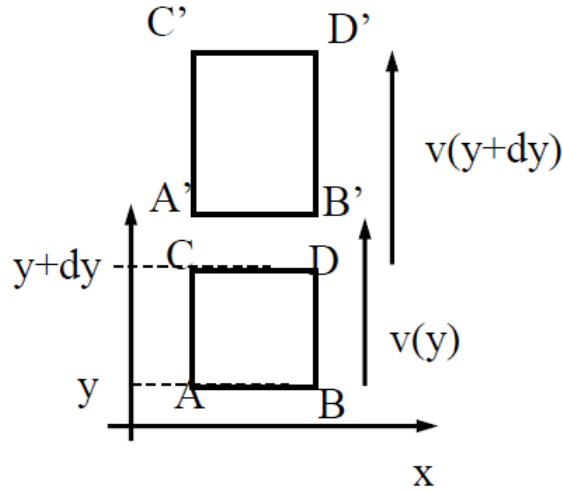


Figure 3.2: Non rigid motion in y direction only

In this case, the fibres in x aligned with the x axis do not change their length, whereas the fibres aligned with the y axis are stretching/compressing. We define the normal strain in y direction ε_y as

$$\varepsilon_y = \frac{\Delta \ell_y}{\ell_y} \quad (3.5)$$

it follows:

$$\varepsilon_y = \frac{\overline{A'C'} - \overline{AC}}{\overline{AC}} = \frac{v(y+dy) - v(y)}{dy} = \frac{v(y) + \frac{\partial v}{\partial y} dy - v(y)}{dy} = \frac{\partial v}{\partial y} \quad (3.6)$$

3.2 Shear strain

We consider now the generic non-rigid displacement defined by (2.1). In this case, the rectangle ABCD is transformed in an irregular quadrilateral A'B'C'D' as show in figure 3.3. In this case, we note that the fibres originally aligned with the coordinate axes change their orientation, so that angles γ_1 and γ_2 become non-zero.

We define the *shear strain* γ_{xy} defined as:

$$\gamma_{xy} = \gamma_1 + \gamma_2 \quad (3.7)$$

with:

$$\begin{aligned}\gamma_1 &= \frac{v(x+dx, y) - v(x, y)}{dx} = \frac{\partial v}{\partial x} \\ \gamma_2 &= \frac{u(x, y+dy) - u(x, y)}{dy} = \frac{\partial u}{\partial y}\end{aligned}\quad (3.8)$$

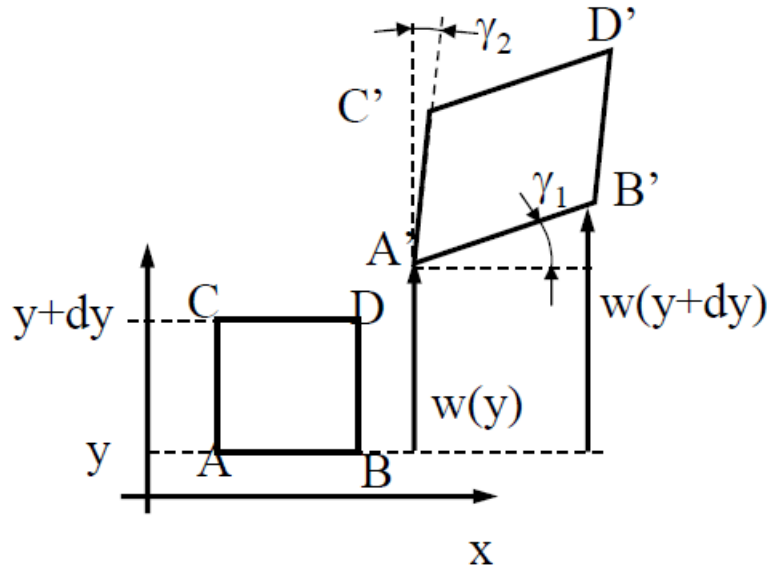


Figure 3.2: Generic planar non-rigid motion

Replacing (3.8) in (3.7) brings:

$$\gamma_{xy} = \frac{\partial u}{\partial y} + \frac{\partial v}{\partial x} \quad (3.9)$$

We recognise that the shear strain defined here is a measure of the relative rotation of the fibres, whereas the normal strains ε_x and ε_y defined previously are a measure of the change in length of the fibres. It can be shown (not reported here for the sake of brevity) that for a 2D case the change in length of any generic fibre not directed along the coordinate axes can be expressed based on the ε_x , ε_y and γ_{xy} strain components. Hence, for the 2D case, the non-rigid component of displacement is fully defined by these three strain components.

3.3 Generalisation to the 3D case.

For a generic three-dimensional solid, we need to define a third component of displacement w along the z coordinate axis. The three displacement components will be in general a function of the three coordinates x, y, z . Hence, the non-rigid displacement of the body is defined as:

$$\begin{aligned}u &= u(x, y, z) \\ v &= v(x, y, z) \\ w &= w(x, y, z)\end{aligned}\quad (3.10)$$

In this case, it is possible to define three components of normal strain, describing the change in length of fibres initially laying along the x, y and z directions:

$$\varepsilon_x = \frac{\partial u}{\partial x} \quad ; \quad \varepsilon_y = \frac{\partial v}{\partial y} \quad ; \quad \varepsilon_z = \frac{\partial w}{\partial z} \quad (3.11)$$

Furthermore, it is possible to define three shear strain components, describing changes in the relative angle of fibres initially laying parallel to the x-y, y-z and z-x directions:

$$\gamma_{xy} = \frac{\partial u}{\partial y} + \frac{\partial v}{\partial x} \quad ; \quad \gamma_{yz} = \frac{\partial v}{\partial z} + \frac{\partial w}{\partial y} \quad ; \quad \gamma_{zx} = \frac{\partial w}{\partial x} + \frac{\partial u}{\partial z} \quad (3.12)$$

4 Stress components

We go back to the 2-D case, considering a solid body consisting of a thin plate having a generic shape in the x-y coordinate plane and having constant thickness s in z direction, as shown in Figure 4.1. From this solid, we extract a prismatic portion having size dx - dy in the x-y plane and thickness s . Next, we consider the forces applied by the rest of the solid on the faces of the prism whose normal is pointing along the x and y axes. Here we consider the normal direction to point from inside to outside the prism, so the faces we focus on are the one on the right and top sides of the dx - dy rectangle in the planar view of Figure 4.1.

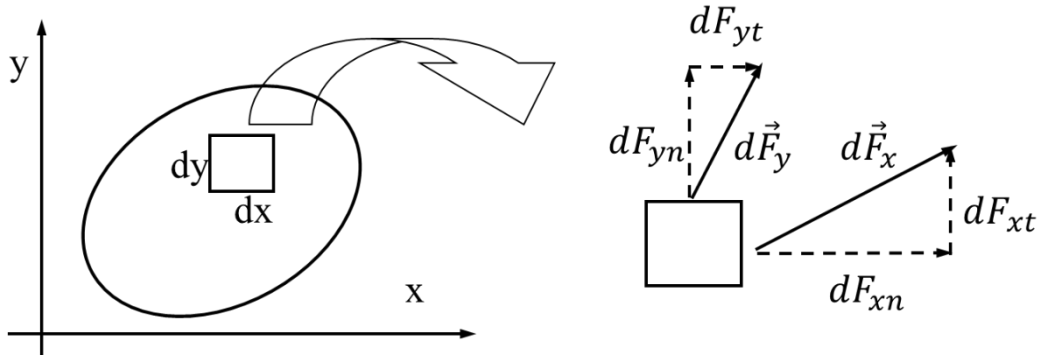


Figure 4.1: A thin plate in the x-y plane

Because the faces of the prism have infinitesimal size, the forces applied on these faces are also infinitesimal. We decompose the total force $d\vec{F}_x$ transmitted through the face having normal pointing in the x direction into a normal component dF_{xn} and a tangential component dF_{xt} and we define the *normal stress* σ_x as the ratio of the normal force component dF_{xn} over the area of the face and the *tangential stress* or *shear stress* on the same face τ_{xy} as the ratio of the tangential force component dF_{xt} over the area of the face:

$$\sigma_x = \frac{dF_{nx}}{sdy} \quad ; \quad \tau_{xy} = \frac{dF_{tx}}{sdy} \quad (4.1)$$

In the same way, for the face whose normal direction points along the y axis we define the normal and tangential stress components, σ_y and τ_{yx} respectively, as:

$$\sigma_y = \frac{dF_{ny}}{sdx} \quad ; \quad \tau_{yx} = \frac{dF_{ty}}{sdx} \quad (4.2)$$

To sum up, a stress component is the ratio of a force over a surface, hence its units in the IS are $\text{N/m}^2 = \text{Pa}$. This ratio is taken over a surface having infinitesimal size, so that it is possible to account for a non-uniform distribution of stresses in a solid. We also distinguish a normal from a tangential stress component, with the normal stress being referred to the component of the force transmitted through the face that is normal to the face itself, while the tangential stress component comes from a component of the force laying in the plane of the face.

The notation used in (4.1) and (4.2) is worth some further clarification. For a normal stress component, we use one single subscript to identify the orientation of the face for which the stress component is computed, the subscript denoting the direction of the normal to the face (according to the sign convention stated above). For a tangential stress component we need however to introduce two subscripts, the first one denoting the normal to the face and the second one denoting the coordinate axis along which the tangential force component is directed. The second subscript is not strictly needed in the planar case, but needs to be introduced when we extend the definitions of normal and tangential stress to the 3D case.

To this aim, we consider a generic 3D solid and within it an infinitesimal prism with edges dx - dy - dz parallel to the coordinate axes and size. In this general case we may define three stress components for each face of the prism. For the face whose normal points along the x axis we have:

$$\sigma_x = \frac{dF_{nx}}{dydz} \quad ; \quad \tau_{xy} = \frac{dF_{bxy}}{dydz} \quad ; \quad \tau_{xz} = \frac{dF_{bxz}}{dydz} \quad (4.3a)$$

In the same way, for the faces with normal pointing along the y and z axes we have:

$$\sigma_y = \frac{dF_{ny}}{dxdz} \quad ; \quad \tau_{yx} = \frac{dF_{byx}}{dxdz} \quad ; \quad \tau_{yz} = \frac{dF_{byz}}{dxdz} \quad (4.3b)$$

$$\sigma_z = \frac{dF_{nz}}{dxdy} \quad ; \quad \tau_{zx} = \frac{dF_{bzx}}{dxdy} \quad ; \quad \tau_{zy} = \frac{dF_{bzy}}{dxdy} \quad (4.3c)$$

Looking again at Figure 4.1 we notice that we might also define some stress components for the face of the prismatic hole produced in the solid by extracting the prism. In particular, if we consider the face on the right side of the prismatic hole, we recognise that the normal to this face is pointing along $-x$ (that is, the normal is parallel to the x axis but points leftwards). On account of Newton's third law, we recognise that the force on this face, which is denoted as $d\vec{F}_{-x}$, is opposite to $d\vec{F}_x$. It follows:

$$\sigma_{-x} = -\sigma_x \quad ; \quad \tau_{-xy} = -\tau_{xy} \quad ; \quad \tau_{-xz} = -\tau_{xz} \quad (4.4a)$$

and for the same reason:

$$\sigma_{-y} = -\sigma_y \quad ; \quad \tau_{-yx} = -\tau_{yx} \quad ; \quad \tau_{-yz} = -\tau_{yz} \quad (4.4b)$$

$$\sigma_{-z} = -\sigma_z \quad ; \quad \tau_{-zx} = -\tau_{zx} \quad ; \quad \tau_{-zy} = -\tau_{zy} \quad (4.4c)$$

Finally, for the simplified planar case, we consider the complete set of forces acting on the infinitesimal prism, as shown in Figure 4.2. In this figure the relationships expressed by (3.16a,b) are taken into account since on the left and bottom face of the prism the normal and tangential forces are considered to have the same magnitude and opposite direction than those applied on the right and top faces respectively.

In a static case, the system of forces needs to satisfy the equilibrium conditions and, in particular, the resultant moment of all forces with respect to the central point C in the prism must be zero. The following equation is obtained:

$$M_c = \tau_{xy} s dy \frac{dx}{2} - \tau_{yx} s dx \frac{dy}{2} + \tau_{xy} s dy \frac{dx}{2} - \tau_{yx} s dx \frac{dy}{2} = 0 \quad (4.5)$$

which brings, after trivial treatment:

$$\tau_{xy} = \tau_{yx} \quad (4.6a)$$

Extending this result to the general 3D case, it is easy to show that the following relationships also hold:

$$\tau_{yz} = \tau_{zy} \quad ; \quad \tau_{zx} = \tau_{xz} \quad (4.6b)$$

Equations (4.6a,b) are known as the *symmetry property of the stress tensor*, given that the nine scalar stress components defined by (4.3a,b,c) can be organised in a 2nd order tensor, i.e. a 3 by 3 matrix, which is symmetric on account of (4.6a,b).

In a dynamic case, based on D'Alembert's principle we should rewrite (4.5) considering the additional term due to the moment of the inertia forces with respect to the centre C. However it can be shown that this is an infinitesimal term of higher order than the other terms in (4.5). Therefore the symmetry property (4.6a,b) is preserved also in a dynamic case.

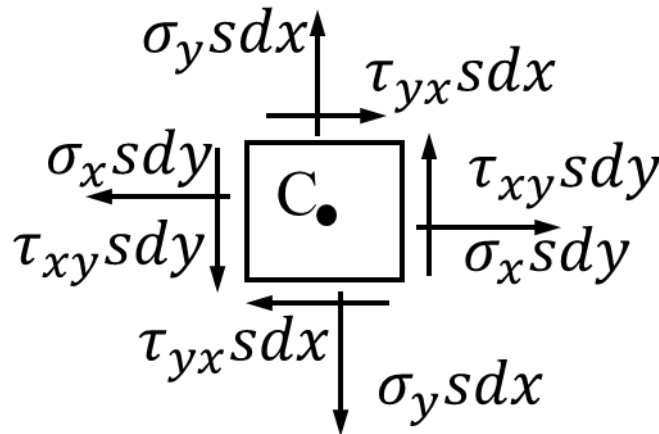


Figure 4.2: Forces acting on an infinitesimal portion of the thin plate

5 Material behaviour – constitutive law

When a solid body is subjected to a system of forces, a field of stresses arises in the body as a consequence. The stress field in turn gives rise to strains, eventually resulting in a non-rigid field of displacements. Knowing the stress-strain relationship for the body is hence pivotal to describing its motion under a given excitation. This relationship is often called *constitutive law* and depends on the mechanical properties of the material out of which the body is made.

A detailed description of the different constitutive laws that may arise for different materials such as metals, elastomers, composites etc. is totally out of the scope of these notes, we will instead restrict our analysis to metallic materials such as steel or aluminium. Experiments show that the behaviour of these materials tends to be isotropic, i.e. the stress-strain relationship does not depend on the direction along which the stresses are applied in the material.

Although the stress-strain relationship involves 6 independent stress components and the same number of strain components, we will start by considering a 1D case where we consider one single component for stress and strain. This is, in first approximation, the case of a tensile test: a specimen of the material under test having axial-symmetric shape is subjected to a stress state by applying two opposite forces at its ends, aligned with the axis of the specimen. The force applied is measured together with the elongation of the central portion of the specimen. From this test, a relationship between the normal stress σ along the specimen axis and the normal strain ε along the same direction is derived. The result is typically a non-linear stress-strain relationship with hysteresis, such as the one shown qualitatively in the upper plot of Figure 5.1: during the loading phase the resulting stress will be larger for the same strain than it is during the unloading case, resulting in a hysteresis cycle and therefore in a mechanism of energy dissipation. For metallic materials, hysteresis effects are small, provided that the maximum stress applied does not exceed the yield

strength. Hence it is often possible to neglect the difference between the upper and lower branch of the hysteresis cycle and describe the stress-strain relationship in terms of a single curve. By doing so, dissipation effects are neglected resulting in an elastic (i.e. conservative) stress-strain relationship. This is in general represented by a non-linear stress-strain relationship as the one shown in the central plot of Figure 5.1. However, it is generally observed that the $\sigma - \varepsilon$ curve is well approximated by a linear relationship, provided that the maximum stress applied does not exceed a limit value defined as the *proportionality limit*. Because most mechanical systems are designed to keep the stress values well below the proportionality limit, we may assume a linear stress-strain relationship in the form:

$$\sigma = E\varepsilon \quad (5.1)$$

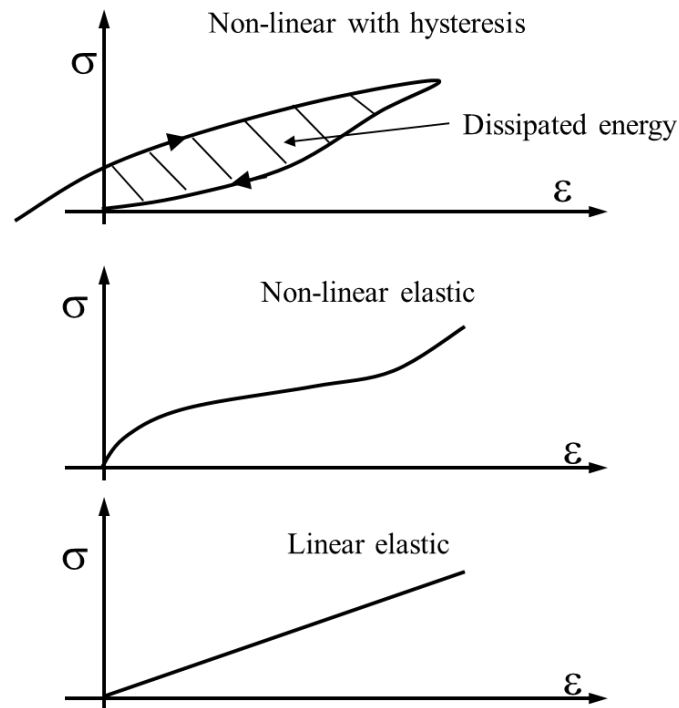


Figure 5.1: Different types of constitutive law

Note that the linear elastic constitutive law (5.1) neglects the energy dissipation effects related with material hysteresis. The use of a different, non-conservative, constitutive law would however result in a huge over-complication of our study. Therefore, we will use in our study the simple linear elastic law and take into account energy dissipation by means of simplified damping models that will be introduced later in the course.

Until now, constitutive laws describing material behaviour were introduced considering only one stress and strain component. However, in reality there are 6 independent strain components and 6 independent stress components. If we confine ourselves to the case of the isotropic linear elastic constitutive law, the complete constitutive law involving all stress and strain components reads as follows:

$$\begin{aligned}
\varepsilon_x &= \frac{1}{E} [\sigma_x - \nu(\sigma_y + \sigma_z)] \\
\varepsilon_y &= \frac{1}{E} [\sigma_y - \nu(\sigma_z + \sigma_x)] \\
\varepsilon_z &= \frac{1}{E} [\sigma_z - \nu(\sigma_x + \sigma_y)] \\
\gamma_{xy} &= \frac{\tau_{xy}}{G} \\
\gamma_{yz} &= \frac{\tau_{yz}}{G} \\
\gamma_{zx} &= \frac{\tau_{zx}}{G}
\end{aligned} \tag{5.2}$$

with E (expressed in $\text{N/m}^2 = \text{Pa}$) the *elastic modulus* or *Young's modulus*, ν the *Poisson's ratio* and G the *shear modulus*. These three parameters are specific properties of a material and can be obtained from material characterisation tests such as a tensile test. For steel material:

$$E = 206000 \text{ N/mm}^2 = 206000 \text{ MPa}$$

$$\nu \cong 0.28 \text{ (pure number)}$$

$$G \cong 80000 \text{ MPa}$$

It should be finally noted that these three parameters are related by:

$$G = \frac{E}{2(1 + \nu)} \tag{5.3}$$

so that the isotropic, linear elastic behaviour of the material can be fully expressed by two of them.

6 Vibrations of beam systems

The topic of vibration of distributed parameter systems will be treated in this course mostly with reference to simple beams and systems of beams. A beam is a deformable body for which one dimension is predominant over the other two. It may be then considered as generated by the extrusion of a given plane shape (called the beam section) along a line or curve in space which is called the axis of the beam. The beam section may have constant or variable shape and size. We will focus our study on beams with rectilinear axis and constant section.

6.1 Internal forces in a beam

The reason why a beam can be studied in simpler terms than a deformable body having generic geometry is that the deformable motion of the beam can be described assuming that each section of the beam moves according to a rigid body motion. In this way, the overall non-rigid motion of the system is decomposed in an infinite number of rigid body motions, one for each section of the beam. We choose the axis of the beam as the x axis of our coordinate reference, so that the motion of the beam can be expressed in terms of the independent coordinates of each beam section, expressed as function of the x coordinate (spatial coordinate) and of time.

Because each section is considered as a rigid body and because for a rigid body static/dynamic equilibrium only depends on the resulting force and moment of a system of forces acting on the body, it also makes sense to introduce the total force and moment components resulting from the distribution of stresses acting on the section. These are called the *internal force* components.

If we consider a beam such as the one represented in Figure 1 and only consider the motion of the beam in the plane of the figure, we may operate a ‘cut’ at some section in the beam so that the system is divided in two trunks. Each trunk exerts on the other one a system of forces that can be reduced to one resulting force and one resulting moment. The force, in turn, can be decomposed in an *axial force* N aligned with the beam axis and a *shear force* T , orthogonal to the beam axis. The resulting moment is known as *bending moment*. The normal and shear force and the bending moment overall form the three internal force components for the beam in a plane. We note in Figure 6.1 that on account of Newton’s third law, the internal force components applied by trunk 1 on trunk 2 are opposite to those applied by trunk 2 to trunk 1.

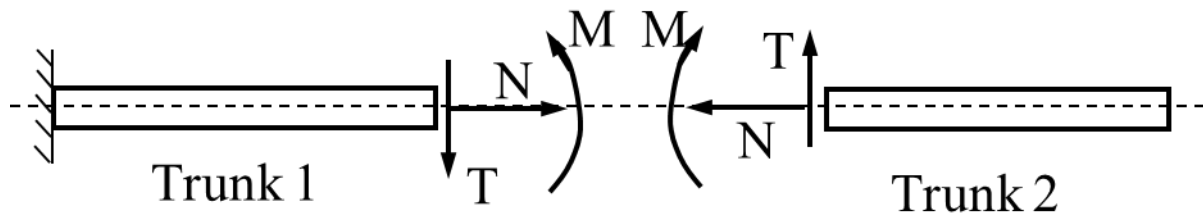


Figure 6.1: Internal forces in a beam with rectilinear axis. 2D case

Before closing this section, it is useful to introduce a set of sign conventions for the internal forces. We do so by taking opposite sign conventions on faces having the beam at their left side (such as the right end of trunk 1) and on their right side (such as the left end of trunk 2). In this way, Newton’s third law is implicitly satisfied. The sign conventions for the internal forces are summarised in Figure 6.2: we notice that axial force N is positive when it produces a traction of the beam, whereas a compressive normal force corresponds to a negative value.

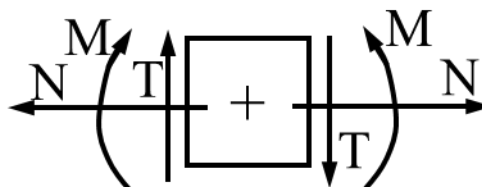


Figure 6.1: Sign conventions for the internal forces in a beam

6.2 Different modes of deformation for a beam with straight axis

Three main modes of deformation are possible for a beam:

- axial deformation: in this mode of deformation, all material points in the beam are moving only in the axial direction (if small contraction/expansion motions of the sections produced by the effect of Poisson’s ratio are neglected). Furthermore, all points on a same section are displacing by the same amount, so that the motion of the beam may be seen as produced by rigid translations in axial direction of all sections. In this mode of deformation, the only internal force generated is the axial force N .
- bending: in this mode of deformation, the beam sections undergo a rigid motion that is the combination of a displacement of the section centre in a direction perpendicular to the axis (what will be called a *transversal displacement*) and of a rotation of the section. In this mode of deformation, a shear force and a bending moment generally arise, whereas the normal force is zero.
- torsion: in this third mode of deformation the sections of the beam rotate around the beam axis (additionally, except in the case of polar symmetry of the beam section, a warping of the

section takes place). Note this mode of deformation does not correspond to a planar motion of the beam. Torsional deformation is accompanied by the arising of shear stresses in the sections of the beam that result in a torsion moment M_t . This is one additional internal force component, not introduced in Figure 6.1. as this figure is only valid for the planar case. More information about torsion will be provided in a next lesson.

Under rather general conditions (and in particular for sections having two axis of symmetry), the three above-described modes of deformation are de-coupled one from the other, which means e.g that bending does not produce elastic or inertial forces affecting the torsion of the beam or vice-versa. This means that the vibrations of the beam involving each of the three modes of deformation may be studied separately, and the overall beam motion may be obtained as the superposition of the three elementary cases represented by the axial vibrations, bending vibrations and torsional vibrations.

7 Glossary

Strain	deformazione
- normal strain	dilatazione
- shear strain	scorrimento
Stress	sforzo
- normal stress	sforzo normale / tensione
- shear stress	sforzo di taglio
Constitutive law	legame costitutivo
Tensile test	prova di trazione
yield strength	tensione di snervamento
beam	trave
Young's modulus	Modulo di Young, modulo di elasticità normale
Poisson's ratio	coefficiente di Poisson
Shear modulus	Modulo di elasticità tangenziale